Week 01 - Descriptives

Action: Environment setup, load dataset, read GitSkills documentation closely **Outcome:** A working, queryable environment with basic descriptives for data

Dataset: GitSkills July 2026 Snapshot

- Research Paper DOI: 10.48550/arXiv.2608.10906
- Dataset DOI: 10.5281/zenodo.21875637

Data Source:

- Hugging Face Parquet Mirror: mvaccargiu/gitskills
- commit: ebab17454a7c236f8f26b183567f1a126f42e3f8

1. Database and Notebook Setup

In [1]:
```
from pathlib import Path
import duckdb
import pandas as pd
import matplotlib.pyplot as plt

ROOT = Path.cwd().parent          # notebooks/ -> project root
RESULTS = ROOT / "results"
FIGURES = ROOT / "figures"
RESULTS.mkdir(exist_ok=True)
FIGURES.mkdir(exist_ok=True)

con = duckdb.connect(str(ROOT / "data" / "gitskills.db"), read_only=True)
```

2. Data Loaded

Done to verify the same quantities of data are loaded as described here. Purposely leaving out artifact siblings as current work doesn't require that data, and it is a large portion of data that would slow research.

In [2]:
```
con.sql("""
    SELECT 'artifacts' AS table_name, count(*) AS n_rows FROM artifacts
    UNION ALL SELECT 'repos', count(*) FROM repos
    UNION ALL SELECT 'mining_runs', count(*) FROM mining_runs
""").df()
```

Out[2]:

	table_name	n_rows
0	artifacts	3797117
1	repos	282200
2	mining_runs	7

3. Overview of Tables

This is the overall layout of key tables within the GitSkills dataset. It includes column names and datatypes for each table.

In [3]:
```
con.sql("DESCRIBE repos").df()[["column_name", "column_type"]]
```

Out[3]:

	column_name	column_type
0	full_name	VARCHAR
1	owner	VARCHAR
2	stars	BIGINT
3	forks	BIGINT
4	is_fork	BIGINT
5	language	VARCHAR
6	license	VARCHAR
7	description	VARCHAR
8	created_at	VARCHAR
9	pushed_at	VARCHAR
10	metadata_fetched	BIGINT

In [4]:
```
con.sql("DESCRIBE artifacts").df()[["column_name", "column_type"]]
```

Out[4]:

	column_name	column_type
0	repo_full_name	VARCHAR
1	path	VARCHAR
2	filename	VARCHAR
3	location_class	VARCHAR
4	file_sha	VARCHAR
5	discovered_at	VARCHAR
6	content	VARCHAR
7	content_fetched	BIGINT
8	frontmatter_valid	BIGINT
9	name	VARCHAR
10	description	VARCHAR
11	body_chars	BIGINT
12	history_fetched	BIGINT
13	composition_fetched	BIGINT
14	dedup_primary	BIGINT
15	first_commit_at	VARCHAR
16	last_commit_at	VARCHAR
17	commit_count	BIGINT
18	sibling_count	BIGINT
19	sibling_bytes	BIGINT
20	has_scripts	BIGINT
21	has_references	BIGINT
22	content_sha_ok	BIGINT
23	composition_truncated	BIGINT
24	first_commit_author	VARCHAR
25	first_commit_author_type	VARCHAR
26	first_commit_message	VARCHAR
27	last_commit_author	VARCHAR
28	last_commit_author_type	VARCHAR
29	last_commit_message	VARCHAR

In [5]:
```
# Temporarily remove col width limits to display full text
with pd.option_context('display.max_colwidth', None):
    display(con.sql("SELECT * FROM mining_runs").df())
```

	run_id	artifact_type	query	started_at	finished_at
0	1	agent-skills	filename:SKILL.md	2026-07-09T17:26:45.768703+00:00	2026-07-09T18:04:34.751737+00:00
1	2	agent-skills	filename:SKILL.md	2026-07-09T18:07:39.150068+00:00	2026-07-12T06:54:51.308396+00:00
2	3	agent-skills	filename:SKILL.md	2026-07-12T06:59:13.601456+00:00	2026-07-20T22:26:56.553671+00:00
3	4	agent-skills	filename:SKILL.md	2026-07-13T12:52:52.309967+00:00	2026-07-20T22:26:56.553671+00:00
4	5	agent-skills	filename:SKILL.md	2026-07-14T18:04:23.050325+00:00	2026-07-20T05:23:31.638481+00:00
5	6	agent-skills	filename:SKILL.md	2026-07-20T05:28:44.667482+00:00	2026-07-20T22:26:56.553671+00:00
6	7	agent-skills	filename:SKILL.md	2026-07-20T07:38:28.581854+00:00	2026-07-20T22:26:56.553671+00:00

4. Profiling Tables

This section analyzes the contents of the repo column currently. More profiling will be done in next weeks report.

In [6]:
```
cols = ["column_name", "column_type", "min", "max", "approx_unique", "q50", "null_p

repos_profile = con.sql("SUMMARIZE SELECT * EXCLUDE (description) FROM repos").df()
repos_profile.to_csv(RESULTS / "week01" / "repos_profile.csv", index=False)

repos_profile[cols].rename(columns={"q50": "median", "approx_unique": "unique_value
```

Out[6]:

	column_name	column_type	min	max	unique_values	median	n
0	full_name	VARCHAR	0-AI-UG/cate	zzzzz1213/Nano-Code-Agent	232901	NaN	
1	owner	VARCHAR	0-AI-UG	zzzzz1213	192218	NaN	
2	stars	BIGINT	0	383040	3520	0	
3	forks	BIGINT	0	80428	1652	0	
4	is_fork	BIGINT	0	1	2	0	
5	language	VARCHAR		xBase	258	NaN	
6	license	VARCHAR		Zlib	44	NaN	
7	created_at	VARCHAR	2008-01-27T17:23:23Z	2026-07-21T02:24:50Z	262796	NaN	
8	pushed_at	VARCHAR	2015-05-08T03:22:57Z	2026-07-21T08:49:38Z	227038	NaN	
9	metadata_fetched	BIGINT	1	1	1	1	

5. Key Details

Some key details I have found, some verifying claims made by dataset creators. This is to support week one's basic descriptives goal.

Copies

This covers how many duplications there are in the dataset. The results of 50.5% match what the dataset paper reports. The intial report is at the following link under "Intended Uses" for the GitSkills dataset

In [7]:
```
con.sql("""
    SELECT
        count(*) AS total_files,
        count(DISTINCT file_sha) AS distinct_contents,
        count(*) - count(DISTINCT file_sha) AS copies,
        round(100.0 * (total_files - distinct_contents) / total_files, 1) AS pct_co
    FROM artifacts
""").df()
```

Out[7]:

	total_files	distinct_contents	copies	pct_copies
0	3797117	1877981	1919136	50.5

Skill Locations

This breaks down where skill files are within projects by individual location.

In [8]:
```
con.sql("""
    SELECT location_class,
        count(*) AS num_files,
        round(100.0 * num_files / sum(num_files) OVER (), 1) AS pct
    FROM artifacts
    GROUP BY location_class
    ORDER BY num_files DESC
""").df()
```

Out[8]:

	location_class	num_files	pct
0	skills-dir	2102053	55.4
1	other	1321412	34.8
2	canonical	373652	9.8

How Widely Are Skills Shared

This counts how many repositories each unique skill appears in, which relates to the core research of whether skills are project specific or widely shared.

In [9]:
```
con.sql("""
    WITH individual_skill AS (
        SELECT file_sha, count(DISTINCT repo_full_name) AS num_repos
        FROM artifacts
        GROUP BY file_sha
    )
    SELECT CASE
        WHEN num_repos = 1    THEN '1 repo'
        WHEN num_repos <= 10  THEN '2-10 repos'
        WHEN num_repos <= 100 THEN '11-100 repos'
        ELSE '100+ repos' END AS shared_across,
    count(*) AS num_skills,
    round(100.0 * num_skills / sum(num_skills) OVER (), 3) AS pct
    FROM individual_skill
    GROUP BY shared_across
    ORDER BY min(num_repos)
""").df()
```

Out[9]:

	shared_across	num_skills	pct
0	1 repo	1571893	83.701
1	2-10 repos	282613	15.049
2	11-100 repos	23005	1.225
3	100+ repos	470	0.025

6. Observations

Loaded Data

- I loaded all artifacts, repos, and mining_runs, and they all matched the published row counts. The data seems to have been collected in three runs between 2026-07-09 and 2026-07-20.

Initial Findings For Broad vs. Project Specific Skills

- Around half of the skill files (50.5%) are copies of another skill file
- Most of the unique skills are not shared, with 83.7% of the skills appearing in only one repository. 470 of the skills appear in more than 100 repositories.
- Copying seems to be concrentrated to a small set of widely shared skills, while most of the skills are written for individual projects.

Skill Locations In Repositories

- 9.8% of files are in the canonical location, or `.claude/skills`
- 55.4% of files live in other recognized skill directories
- 34.8% of files are elsewhere Analyzing skill location may help separate shared skills from project-specific skills.

Repo Table Profiling

- Most skills live in small projects, with the median stars and forks being zero
- Repository age shouldn't be used to determine skill creation date, as repo age goes back to 2008